

Introduction to Statistical Inference

What is a p-value?

A p-value is the probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true. A low p-value (typically < 0.05) suggests strong evidence against the null hypothesis.

Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions about a population based on sample data. It involves a null hypothesis (H_0) and an alternative hypothesis (H_1). The test statistic measures how far the sample data falls from the null hypothesis value.

Type I and Type II Errors

A Type I error (false positive) occurs when we reject a true null hypothesis. Its probability is denoted by α (significance level). A Type II error (false negative) occurs when we fail to reject a false null hypothesis. Its probability is denoted by β . Statistical power = $1 - \beta$.

Confidence Intervals

A confidence interval gives a range of plausible values for a population parameter. A 95% confidence interval means that if we repeated the sampling procedure many times, 95% of the intervals would contain the true parameter.